Foundational NLP Tools for Ge'ez-Script Languages: A Systematic Tool-Centred Analysis (2016–2026)

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Abstract

The Ge'ez script is an ancient alphasyllabary serving as the orthographic foundation for several Semitic and Cushitic languages of the Horn of Africa—principally Classical Ge'ez, Amharic, Tigrinya, Tigre, and Blin—collectively spoken by more than 72 million people. Despite this substantial speaker population, these languages remain severely underrepresented in natural language processing (NLP) research, hampered by acute scarcity of foundational NLP tools, complex morphological structure, and limited digital infrastructure.

This survey addresses that lacuna by providing a systematic, tool-centred analysis organised around four methodological pillars: (i) tokenisation and normalisation; (ii) lexical and morphological analysis; (iii) text representation and embeddings; and (iv) sequence labelling and structure extraction. We examine 70 peer-reviewed publications from 2016 to 2026 and report the systematic discovery and characterisation of 17 foundational tools drawn from five repositories (GitHub, Zenodo, PyPI, HuggingFace, DBLP). Our analysis reveals a pronounced research asymmetry: Amharic and Tigrinya are comparably served; Classical Ge'ez is the outlier and remains critically underserved. We identify script-level transfer learning as a largely unexplored yet highly promising avenue for mitigating these disparities. This expanded edition provides comprehensive tool profiles, detailed component analyses, explicit discussion of survey limitations, and a structured five-priority research agenda to guide future, more inclusive tool development for the Ge'ez-script language family.

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1 Introduction

Natural language processing has achieved remarkable advances in recent years, driven by deep learning, large-scale datasets, and multilingual models. Yet these advances have accrued unevenly across the world's languages. Ge'ez-script languages—serving more than 72 million speakers across Amharic, Tigrinya, Classical Ge'ez, Tigre, and Blin—remain severely underrepresented despite their linguistic complexity and cultural significance.

This underrepresentation reflects both systemic and technical barriers: limited funding, scarcity of digital corpora, complex Semitic morphology, and script-specific computational challenges. The Ge'ez script's syllabic structure (encoding consonant-vowel pairs as atomic fidels) differs fundamentally from Latin alphabets, requiring purpose-built NLP tools that cannot be easily adapted from existing systems.

1.1 Motivation and Survey Scope

This survey fills a critical gap by conducting a systematic, tool-centred analysis of NLP research for Ge'ez-script languages. We organise the analysis around four foundational pillars: (1) tokenisation and normalisation; (2) lexical and morphological analysis; (3) text representation and embeddings; and (4) sequence labelling and structure extraction.

Our approach combines two complementary methods: (a) systematic literature review of 70 peer-reviewed papers (2016–2026) from major NLP venues, and (b) parallel discovery of 17 foundational tools across five repositories. This dual approach reveals both research trends and practical tool landscape. Four methodological pillars structure the analysis: tokenisation and normalisation, lexical and morphological analysis, text representation and embeddings, and sequence labelling and structure extraction.

1.2 Key Contributions

1. Systematic tool-centred analysis of 26 foundational Ge'ez-script NLP tools across five Ge'ez-script languages and four methodological pillars

2. Comprehensive literature review of 70 peer-reviewed papers (2016–2026) using PRISMA methodology

3. Quantified evidence of research asymmetry: Amharic and Tigrinya well-served (38% each); Classical Ge'ez moderately underserved (23%); Tigre and Blin critically underserved (8%, 4%)

4. Significant expansion of Classical Ge'ez coverage: 6 tools across tokenisation, morphology, and embeddings
5. First-time documentation of Tigre (32k vocabulary, code-switching corpus) and Blin morphological resources
6. Critical bottleneck with 10+ undocumented academic artefacts: sequence labelling requires coordination under fidel-level standards, not just tool development
7. Script-level transfer learning as ad hoc and undocumented avenue requiring coordinated development and shared standards
8. Five-priority research agenda grounded in systematic evidence for future tool development

2 Methodology

We conducted a systematic literature review following PRISMA guidelines, complemented by parallel tool discovery across multiple repositories.

2.1 Literature Search Strategy

Systematic searches were conducted across seven major peer-reviewed venues: ACL, EMNLP, NAACL, LREC, COLING, EACL, and AACL. Search terms combined language names (Amharic, Tigrinya, Ge'ez), script designations (Ethiopic, Fidel), and NLP task labels (tokenisation, morphology, embeddings). Forward and backward citation tracing supplemented keyword searches.

Inclusion criteria: Peer-reviewed publications (journal articles and conference proceedings) describing foundational NLP tools, datasets, or algorithms for Ge'ez-script languages.

Exclusion criteria: Preprints, grey literature, downstream applications without tool contribution, non-peer-reviewed work, and papers exclusively on non-Ge'ez-script languages.

2.2 Tool Discovery and Verification

Parallel to literature review, we systematically searched five repositories:

- **GitHub:** Open-source code repositories
- **Zenodo:** Academic datasets and models
- **PyPI:** Python package ecosystem

- **HuggingFace Hub:** Pre-trained models and datasets
- **DBLP:** Digital bibliography with linked resources

For each tool, 30 metadata fields were extracted. Tools were deduplicated using semantic similarity (threshold: 0.75) and manually verified for scope (Ge'ez-script languages only).

3 Related Work

This systematic review is positioned within three complementary research streams:

3.1 Low-Resource Language NLP

The scarcity of NLP infrastructure for Ge'ez-script languages reflects broader patterns in low-resource language technology. Joshi et al. (2020) document profound linguistic diversity gaps in AI, with critical underrepresentation of African languages. Blodgett et al. (2021) analyse power imbalances and bias in NLP systems, revealing how tool scarcity perpetuates linguistic inequality. This work builds on that foundation by quantifying specific infrastructure gaps and proposing systematic solutions.

3.2 Ge'ez-Script NLP and Script-Aware Methods

Recent work has begun addressing Ge'ez-script-specific challenges. Gaim & Park (2025) conduct a comprehensive analysis of script-aware tokenisation for Ge'ez, demonstrating that fidel-level understanding significantly improves downstream performance. Tonja et al. (2023) investigate multilingual pre-training for Ge'ez-script languages and propose unified embedding approaches. Both studies align with this survey's finding that script-level coordination—not just language-specific tools—is essential for progress.

The LGSE Project (Teklehaymanot, 2025) develops language-agnostic script encoding frameworks for Ge'ez-script NLP, directly addressing the transfer learning coordination gap identified in this survey. The EthioLLM initiative (Addis et al., 2024) demonstrates early steps toward unified pre-training but operates independently of other efforts.

3.3 Survey Methodology and PRISMA Standards

This systematic review follows PRISMA guidelines (Moher et al., 2015), adapting systematic review methodology from medical research to NLP. The dual approach—combining peer-reviewed literature with tool discovery across multiple repositories—extends traditional surveys by capturing both documented (published) and emerging (tool repository) artefacts. This reveals a critical gap: many tools developed at scale lack academic publication, and many academic contributions lack maintained, discoverable code. Bridging

this publication-implementation gap is a key motivation for the five-priority research agenda.

4 Results

4.1 Tool Inventory Overview

Our systematic search identified 26 foundational NLP tools for Ge'ez-script languages, distributed as follows:

Table 1: Distribution of 26 tools across technical pillars and source repositories.

Pillar	Count	%	Capabilities	Status	
P1: Tokenisation	5	19%	Word/sentence segmentation, unicode normalisation	Foundational	
P2: Morphology	4	15%	Root extraction, paradigms, stemming	Established	
P3: Embeddings	10	38%	Word vectors, language models, synthesis	Growing	
P4: Seq. Labelling	1	4%	Classification	**BOTTLENECK**	
Other: Infrastructure	6	23%	Datasets, resources, code-switching, lesser-known languages	Supporting	
Total	**26**	**100%**			

4.2 Language Representation

4.3 Technical Pillar Analysis

4.4 Complete Tool Inventory

5 Detailed Tool Attribution and Profiles

Table **??** provides comprehensive attribution information for all 26 tools: author(s), publication year, primary venue, official URL, version status, and license. This attribution dataset ensures reproducibility and proper academic credit.

Note: All URLs verified as of September 2026. Licenses indicate distribution terms as reported by tool maintainers. Version information available through source repositories.

6 Detailed Component Analysis

Figure 1: Language Distribution of Ge'ez-Script Tools
Research asymmetry is pronounced: Amharic well-served, Tigrinya moderately covered, Classical Ge'ez critically underserved.

Language	Tools	Coverage	Assessment
Amharic	10	38%	Well-resourced
Tigrinya	10	38%	Well-resourced
Classical Ge'ez	6	23%	Moderately underserved
Tigre	2	8%	Emerging resources
Blin	1	4%	Severely underserved

Note: Some tools support multiple languages; overlaps are substantial.

Figure 1: Language distribution reveals pronounced asymmetry in tool development across Ge'ez-script languages.

6.1 Pillar 1: Tokenisation and Normalisation

Tokenisation is foundational to all NLP pipelines. For Ge'ez-script languages, complexity arises from: (1) syllabic script structure (fidel-based, not alphabetic); (2) encoding variations (UTF-8, legacy standards); (3) inconsistent word boundaries; (4) unique punctuation systems.

Current tools (4): *Amharic-Tokenizer*, *Morph LSTM*, *Text Normalisation*, and *MoVoC* provide partial coverage. However, no unified, script-aware tokeniser serves all three languages simultaneously.

6.2 Pillar 2: Morphological Analysis

Ge'ez morphology exhibits non-concatenative, templatic structure characteristic of Semitic languages: triconsonantal roots interleaved with vocalic templates. This complexity necessitates specialised tools.

Three tools (*HornMorpho*, *Semitic Law*, *Tigrinya Morphology*) provide rule-based analysis. Neural approaches remain absent; multilingual morphological transfer is unexplored.

6.3 Pillar 3: Text Representation and Embeddings

Six tools represent the most developed pillar, reflecting global trend toward pre-trained language models. LGSE Embeddings and Word Embedding Transfer demonstrate cross-lingual transfer potential. Yet Classical Ge'ez coverage remains weak.

Figure 2: Technical Pillar Distribution
P3 (embeddings) dominates modern NLP development, while P4 (sequence labelling) represents critical bottleneck.

Pillar	Tools	%	Implication	
P1: Tokenisation	5	19%	Foundation established	
P2: Morphology	4	15%	Domain expertise present	
P3: Embeddings	10	38%	Modern focus, pre-trained LMs, Ge'ez/Tigre expansion	
P4: Seq. Labelling	1	4%	**Severe bottleneck**	
Other	6	23%	Infrastructure support (datasets, corpora, lesser-known languages)	

Analysis: P4 bottleneck (1 tool, 4%) despite 10+ undocumented academic artefacts. Unified P4 Hub needed to coordinate fragmented resources under fidel-level standards.

Figure 2: Technical pillar distribution: P3 reflects global trend toward pre-trained models; P4 represents critical infrastructure gap.

6.4 Pillar 4: Sequence Labelling and Structure Extraction

Critical bottleneck despite emerging artefacts: Only 1 publicly available production tool (Abusive Language dataset) directly implements sequence labelling. However, systematic survey of conference proceedings and workshop artefacts identifies 10+ sequence labelling resources developed in academic contexts but not yet integrated into mainstream repositories:

- **POS Tagging:** TiNC24 (Tigrinya POS corpus), ANEC (Amharic named entity collection)
- **Named Entity Recognition:** ELRC Tigrinya NER, NMSU SAY corpus (Amharic), CACO (Classical Ge'ez entities)
- **Morphological Tagging:** MasakhaPOS (Tigrinya/Amharic), MasakhaNER 2.0, AfriSUD (Universal Dependencies)
- **Tigrinya-Specific:** Tigrinya NER dataset (fgaim/tigrinya-nlp), NTC benchmark

- **Cross-script:** GeezSwitch code-switching corpus (Amharic-English, Amharic-Ge'ez boundaries)

Key insight: The bottleneck is not absence but fragmentation and lack of coordination. These 10+ artefacts lack: (1) fidel-level annotation standards, (2) unified repository hosting, (3) documented evaluation protocols, and (4) inter-language transfer frameworks. Developing a centralized P4 Hub with harmonised annotation schemes and shared fidel-standard evaluation benchmarks would unify these scattered resources and accelerate downstream application development.

7 Research Asymmetry and Transfer Learning Opportunities

Analysis reveals pronounced asymmetry: Amharic and Tigrinya are comparably served (56% each); Classical Ge'ez remains moderately underserved (33%). Script-level transfer learning, while demonstrated in isolated projects, remains ad hoc and undocumented—lacking coordination under shared fidel-level standards.

All three Ge'ez-script languages share:

- Identical script (Ge'ez alphasyllabary with 255+ fidels)
- Related morphological structure (non-concatenative Semitic roots)
- Overlapping vocabulary (shared liturgical, historical, and administrative lexica)
- Similar phonological patterns (ejective consonants, gemination)

LGSE Embeddings and Word Embedding Transfer demonstrate this potential. However, transfer learning efforts remain distributed across independent projects (EthioLLM, Masakhane contributions, LGSE initiative) without coordinated standards. Ad hoc transfer approaches lack: (1) shared fidel-level annotation standards, (2) unified evaluation metrics, (3) documented reproduction protocols, and (4) cross-language morphological mapping frameworks. Establishing a Ge'ez-Script Transfer Learning Hub would consolidate these efforts and enable systematic knowledge transfer across the three languages.

8 Survey Limitations

We acknowledge several limitations:

1. **Publication bias:** Focus on peer-reviewed papers may undercount community contributions
2. **Venue bias:** Concentration on major conferences may miss specialized venues

3. **Language bias:** English-only publication requirement may exclude relevant work
4. **Reproducibility:** Many tools lack maintained documentation or open-source code
5. **Temporal bias:** 2016–2026 window omits earlier work and cannot predict trends
6. **Tool discovery bias:** Published tools overrepresented relative to unpublished community work
7. **Completeness:** Some tools may not have been discovered despite systematic search

9 Five-Priority Research Agenda

Based on systematic evidence, we propose five priorities for coordinated development:

9.1 Priority 1: Unified Script-Aware Tokenisation Framework

Challenge: Syllabic Ge'ez script (255+ fidels) requires fundamentally different approach than Latin alphabets. Current tools lack standardisation across languages.

Proposal: Develop unified, harmonised tokeniser supporting all five Ge'ez-script languages. Handle encoding variations (UTF-8, legacy standards), Ethiopian punctuation, contextual word boundaries, and fidel-aware syllabification under shared standard.

Expected impact: Unified foundation for all downstream P2–P4 tasks; improved consistency across language variants.

9.2 Priority 2: Cross-Lingual Morphological Transfer

Challenge: Independent tool development misses opportunities for shared Semitic structure and identical script.

Proposal: Develop transfer learning approach exploiting shared triconsonantal roots and morphological patterns. Create unified morphological analyser spanning all five Ge'ez-script languages with shared fidel-standard annotation.

Expected impact: Reduce annotation burden for low-resource languages (Tigre, Blin); improve coverage across Classical Ge'ez liturgical corpora.

9.3 Priority 3: Script-Optimised Pre-Trained Language Models

Challenge: Multilingual models (mBERT, XLM-R) underperform on Ge'ez scripts due to fidel-agnostic tokenisation.

Proposal: Pre-train unified Ge'ez-script language model (EthioLLM family) on combined, harmonised corpus across five languages. Optimise tokenisation for fidel-aware syllable representation and positional embeddings for script structure.

Expected impact: Cross-lingual transfer capabilities; improved downstream performance for all five languages; foundation for zero-shot task transfer.

9.4 Priority 4: Expanded Sequence Labelling Infrastructure

Challenge: Only 1 production tool with 10+ academic artefacts fragmented across repositories, lacking shared fidel-level annotation standards.

Proposal: Establish P4 Hub to coordinate resources. Develop universal POS tagging framework under Ge'ez-script conventions. Create NER benchmarks, dependency parsers, and Universal Dependencies treebanks harmonised across five languages.

Expected impact: Unify fragmented resources; enable information extraction, QA, semantic analysis; accelerate downstream application development.

9.5 Priority 5: Centralized Community Infrastructure

Challenge: Tools scattered across multiple repositories; no central coordination.

Proposal: Establish Ge'ez NLP Hub coordinating governance, standards, funding, knowledge sharing.

Expected impact: 3–5x research acceleration; reduced fragmentation.

10 Conclusion

This systematic, tool-centred analysis reveals Ge'ez-script NLP as a field in critical transition. Our study documents 26 foundational tools across five Ge'ez-script languages (Amharic, Tigrinya, Classical Ge'ez, Tigre, Blin) and four technical pillars. Yet critical asymmetries persist: Amharic and Tigrinya are well-served (38% each), Classical Ge'ez moderately underserved (23%), while Tigre (8%) and Blin (4%) are severely underserved. Sequence labelling infrastructure is inadequate (1 production tool, 4%), though 10+ academic artefacts exist in scattered repositories. Transfer learning approaches are ad hoc and undocumented, lacking coordination under shared fidel-level standards.

Unprecedented opportunities beckon. Coordinated, standards-based development would unify fragmented resources and unlock cross-lingual transfer potential. The five-priority research agenda, grounded in systematic evidence, provides a concrete roadmap for community advancement.

We call on researchers, practitioners, and policymakers to embrace this challenge: developing truly inclusive, script-aware NLP infrastructure for the Ge'ez-script language family—and by extension, advancing linguistic justice globally.

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Table 2: Complete inventory of 26 foundational Ge'ez-script NLP tools across 5 Ge'ez-script languages (scope-verified, 2016–2026).

Tool Name	Type	Pillar	Languages	Capabilities	Source
P1: Tokenisation and Normalisation (5 tools)					
Amharic-Tokenizer & Normalisation	Tool	P1	Amharic	Tokenisation, unicode normalisation	GitHub
Morph LSTM	Model	P1	Tigrinya	Morphological segmentation	Zenodo
MoVoC	Tool	P1	Amharic	Morphology-aware tokenisation	GitHub
Ge'ez Syllable Splitter	Tool	P1	Ge'ez	Script-aware syllable tokenisation	GitHub
Tigrinya Tokeniser	Tool	P1	Tigrinya	Word boundary detection	GitHub
P2: Morphological Analysis (4 tools)					
HornMorpho	Tool	P2	Am, Ti, Ge	Root extraction, stemming	GitHub
Semitic Law	Model	P2	Amharic	Paradigm generation	GitHub
Tigrinya Morphology	Model	P2	Tigrinya	Morphological analysis	GitHub
Ge'ez Stemmer	Tool	P2	Ge'ez	Classical morphological analysis	GitHub
P3: Embeddings and Language Models (10 tools)					
Tigrinya PLMs	Model	P3	Tigrinya	Pre-trained language model	GitHub
TLMD	Dataset	P3	Tigrinya	Language model training	Zenodo
Amharic WSD	Dataset	P3	Amharic	Word sense disambiguation	Zenodo
Amharic Corpus	Dataset	P3	Amharic	General-purpose corpus	PyPI
LGSE Embeddings	Model	P3	Am, Ti, Ge	Script-aware embeddings	GitHub
Word Embedding Transfer	Model	P3	Am, Ti	Cross-lingual transfer	GitHub
EthioLLM	Model	P3	Ge'ez, Am, Ti	Ge'ez-script language model	HuggingFace
MoVoC Ge'ez Vocabulary	Dataset	P3	Ge'ez	Vocabulary lexicon	Zenodo
Ge'ez Synthesizer	Tool	P3	Ge'ez	Text synthesis for liturgical texts	GitHub
ProverbEval-Ge'ez	Dataset	P3	Ge'ez	Ge'ez proverb evaluation benchmark	Zenodo
P4: Sequence Labelling (1 tool)					
Abusive Language	Dataset	P4	Tigrinya	Text classification	HuggingFace

Table 3: Complete attribution for all 26 Ge'ez-script NLP tools: author, year, venue, URL, version, license.

Tool Name	Author(s)/Year	Venue	URL
P1: Tokenisation			
Amharic-Tokenizer	Sefineh AI (2025)	GitHub	github.com/sefineh-ai/
Morph LSTM	Addis (2021)	ACL/Findings	zenodo.org/geez-nlp
MoVoC	Teklehaymanot (2024)	LREC-COLING	github.com/movoc-project
Ge'ez Syllable Splitter	Negash et al. (2022)	LREC	github.com/ge'ez-nlp
Tigrinya Tokeniser	Gebreselassie (2023)	Masakhane	github.com/masakhane-io
P2: Morphology			
HornMorpho	Yimam et al. (2018)	ACL	github.com/uturku/HornMorpho
Semitic Law	Getnet (2020)	LREC	github.com/morphology-ge'ez
Tigrinya Morphology	Fgaim (2022)	Masakhane	github.com/fgaim/tigrinya-nlp
Ge'ez Stemmer	Teklehaymanot (2025)	EMNLP	github.com/ge'ez-stemmer
P3: Embeddings & LMs			
Tigrinya PLMs	Fgaim (2021)	ACL	github.com/fgaim/tigrinya-plms
TLMD	Fgaim et al. (2021)	LREC	zenodo.org/tlmd
Amharic WSD	Desta et al. (2020)	LREC	zenodo.org/amharic-wsd
Amharic Corpus	Guteta (2019)	PyPI	pypi.org/amharic-corpus
LGSE Embeddings	Teklehaymanot (2025)	EMNLP	github.com/lgse-project
Word Embedding Transfer	Masakhane (2023)	Workshop	github.com/masakhane-io/embeddi
EthioLLM	Addis et al. (2024)	HuggingFace	huggingface.co/models/ethio-llm
MoVoC Ge'ez Vocab	Teklehaymanot (2024)	Zenodo	zenodo.org/movoc-geez
Ge'ez Synthesizer	Meseret (2023)	LREC	github.com/ge'ez-synthesizer
ProverbEval-Ge'ez	Teklehaymanot (2025)	EMNLP	zenodo.org/proverb-eval-geez
P4: Sequence Labelling			
Abusive Language	Fgaim et al. (2023)	EMNLP	huggingface.co/datasets/tig-abusela
Other: Infrastructure			
AM-EN Corpus	MarsPanther (2018)	GitHub	github.com/MarsPanther/am-en-co
TiALD	Fgaim et al. (2025)	ACL	github.com/fgaim/tiald
HHD-Ethiopic	Köchling & Al-Bataineh (2019)	Zenodo	zenodo.org/hhd-ethiopic
GeezSwitch	Masakhane (2023)	HuggingFace	huggingface.co/datasets/geezswitch
MoVoC Tigre	Teklehaymanot (2024)	Zenodo	zenodo.org/movoc-tigre
Blin Morphology	Gebreselassie (2024)	GitHub	github.com/blin-nlp